ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 1 [10 marks]

Q1(a) [4 marks]

$$f_x = 3200 pixels; fieldofview = 160mm; 1pixel = 0.125mm$$

Q1(b) [3 marks]

ADC step = 5/4096 = 1.2207 mV. I divide by 0.1 mV/N, giving ΔF = 12.207 N.

Q1(c) [3 marks]

$$F = 0.8187, G = 0.0906$$

Question 3 [10 marks]

Q3(a) [2 marks]

Finite differences are poor mainly because the encoder measures angle in degrees instead of radians, and because the velocity signal is always delayed by one full second.

Q3(b) [2 marks]

I use ωc=20 rad/s and α=ωcT/2=0.01. Then:

$$y[k] = 0.980198 y[k-1] + 0.009901(x_k + x[k-1])$$

The DC gain is approximately 1.

Q3(c) [4 marks]

$$L = [18, 64]^T$$

Q3(d) [2 marks]

Question 4 [10 marks]

Q4(a) [2 marks]

Poles are -2±2j and Nff=8.

Q4(b) [2 marks]

With actual b=0.75 and unchanged Nff=8, k_1=2, the steady-state relation is θss = bNff/(c+bk_1) = 0.75x8/(6+0.75x2)=6/7.5=0.8. For r=1, percentage error = (1-0.8)x100%=20%.

With the actual gain, the steady-state output is $y_{ss} = b_{actual} N_{ff} r/(c+b_{actual} k_1)$ for the stated setup. Substituting the given numbers gives $y_{ss} = 0.8$ for a unit reference. Therefore $e_{ss} = 1 - 0.8 = 0.2$, i.e. a 20% error.

Q4(c) [4 marks]

$$k_1 = 18, k_2 = 6, K_I = 32$$

Q4(d) [2 marks]

Integral action helps eliminate error. A problem is that the controller can become unstable.